[Hands-On Activity: Cleaning data with spreadsheets]

Activity overview

You've learned about cleaning data and its importance in meeting good data science standards. In this activity, you'll do some data cleaning with spreadsheets, and then transpose the data.

By the time you complete this activity, you will be able to perform some basic cleaning methods in spreadsheets. This will enable you to clean and transpose data, which is important for making data more specific and accurate in your career.

Follow the instructions to complete each step of the activity. Then answer the questions at the end of the activity before going to the next course item.

To get started, first access the data spreadsheet.

To use the spreadsheet for this course item, click the link below and select “Use Template.”

Link to data spreadsheet: Cleaning with spreadsheets

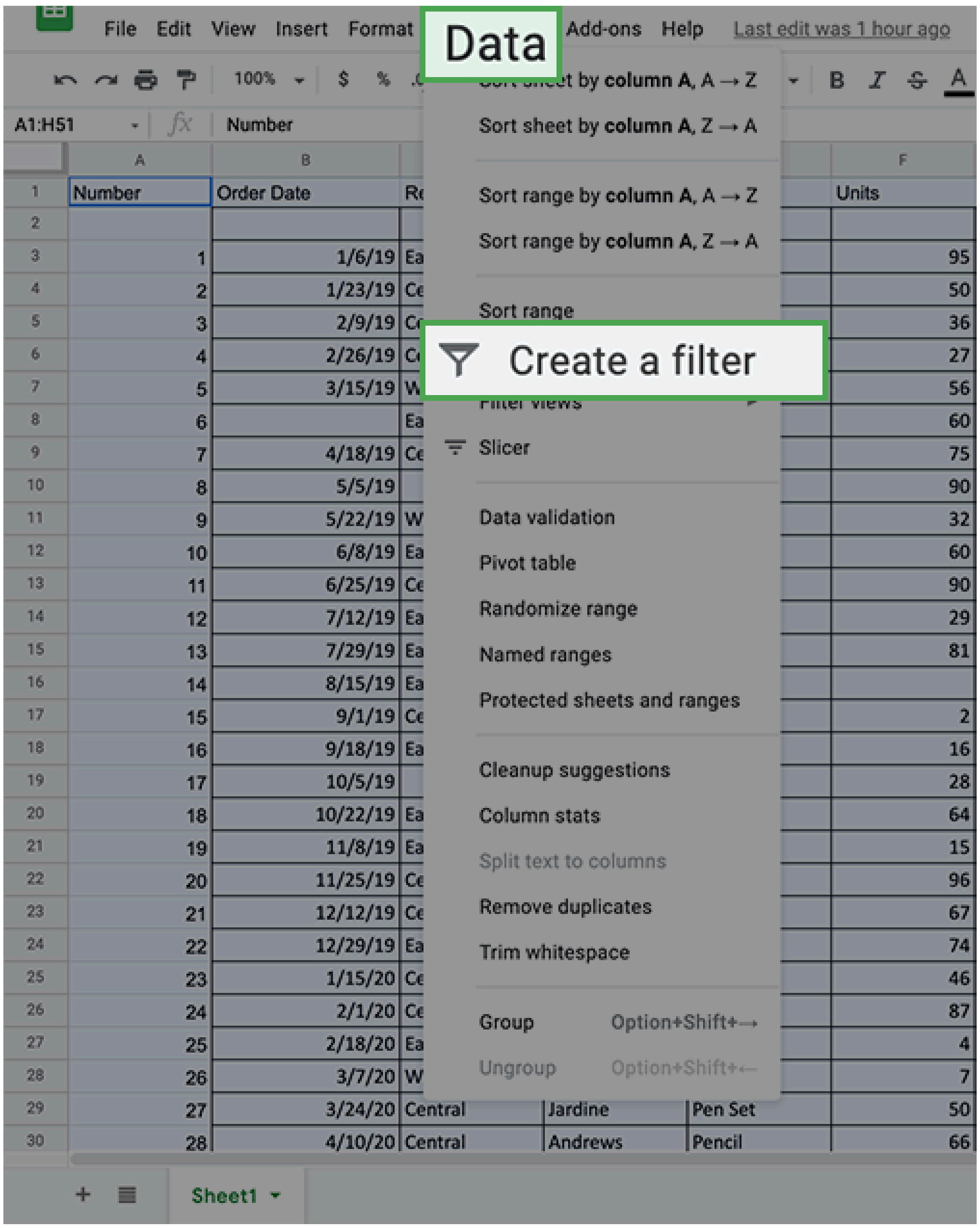
OR

If you don’t have a Google account, you can download the template directly from the attachment below.

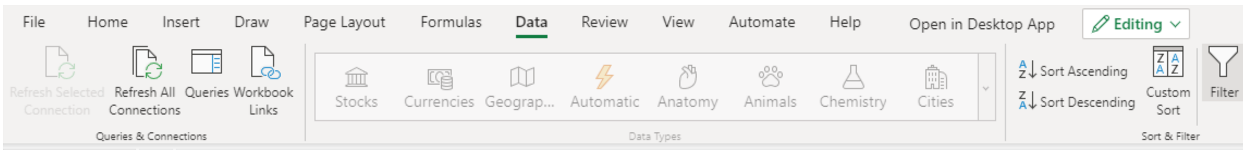
The first technique we’ll use is to select and eliminate rows containing blank cells by using filters. To eliminate rows with blank cells:

1. Highlight all cells in the spreadsheet. You can highlight **Columns A-H** by clicking on the header of **Column A**, holding **Shift**, and clicking on the header of **Column H**.

2. Click on the **Data** tab and pick the **Create a filter** option. In Microsoft Excel, this is called **Filter**.

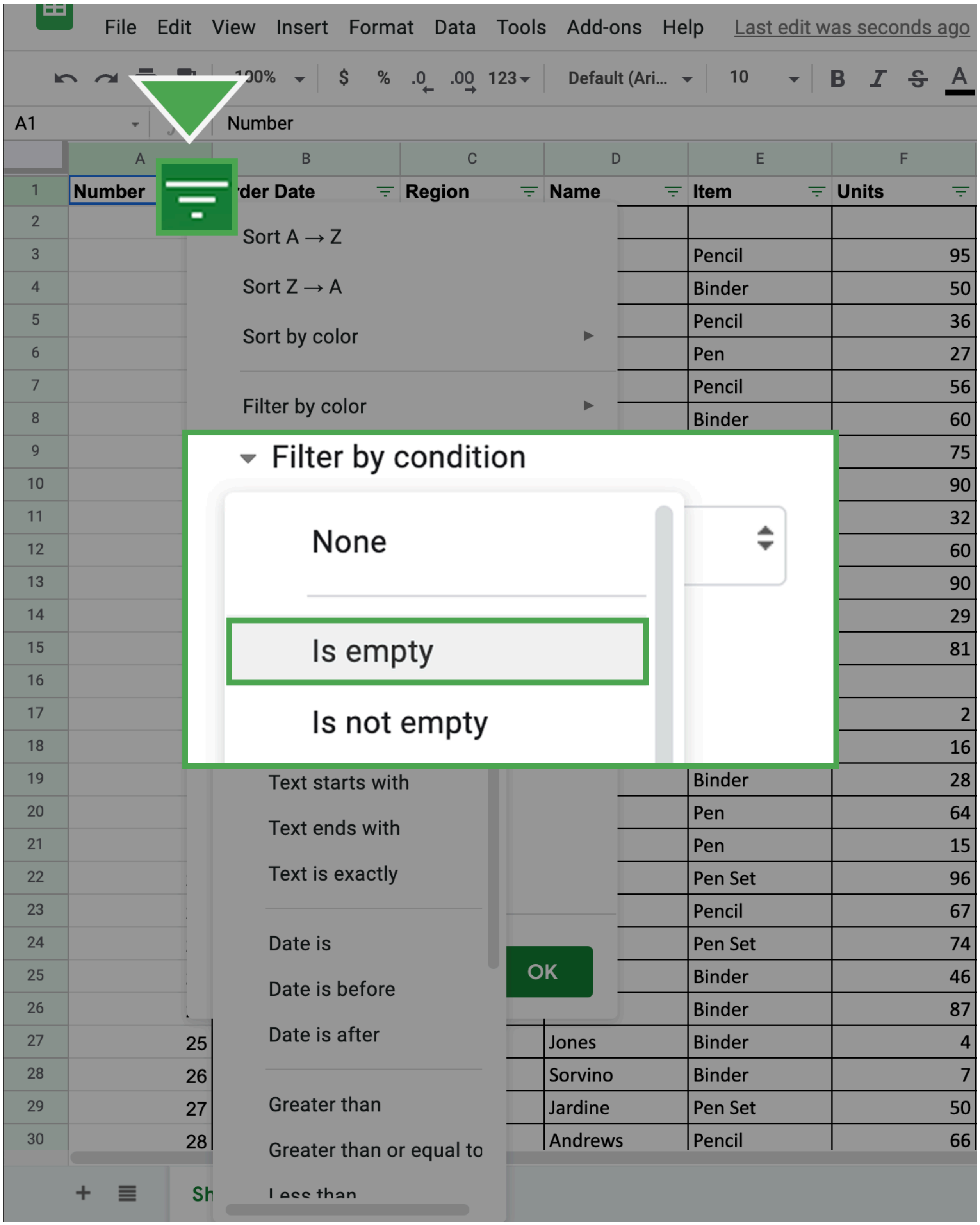


Excel:



3. Every column now shows a green triangle in the first row next to the column title. Click the green triangle in **Column B** to access a new menu.

4. On that new menu, click **Filter by condition** and open the dropdown menu to select **Is empty.** Click **OK**.


File Edit View Insert Format Data Tools Add-ons Help
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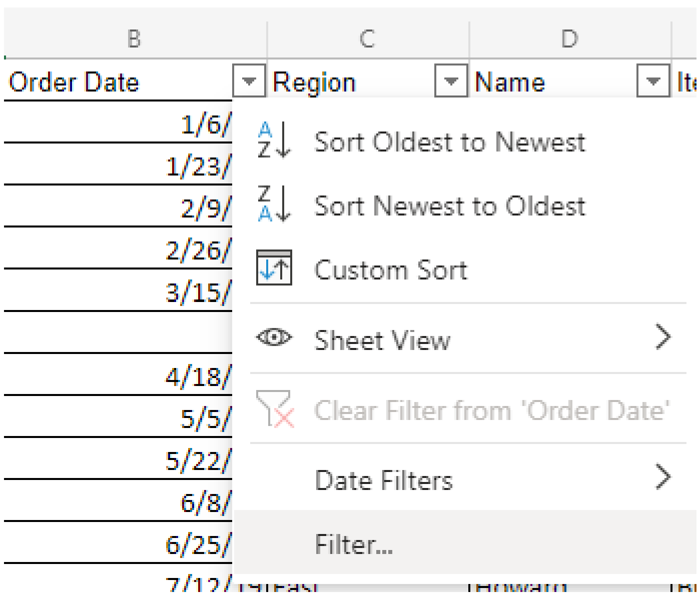
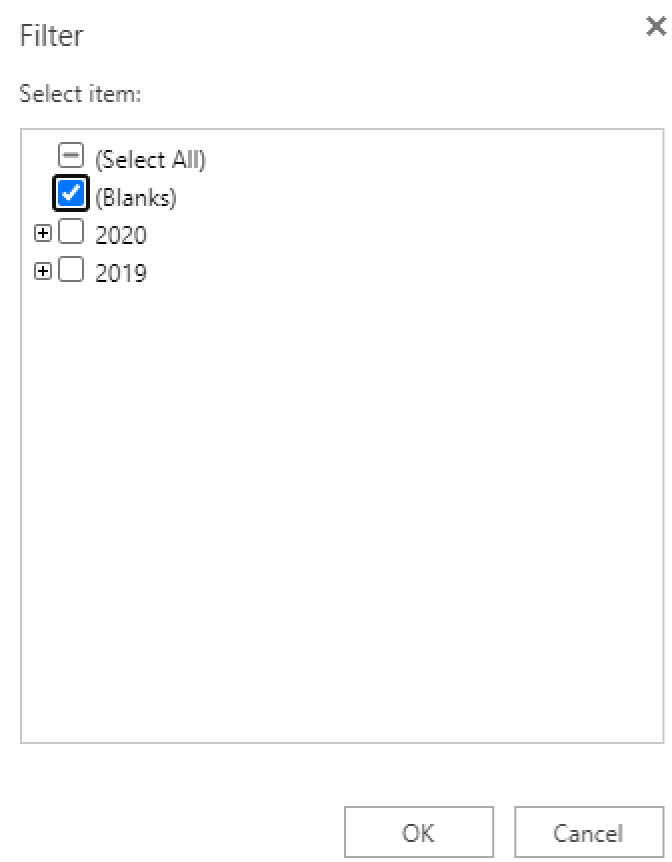
Filter by color
Filter by condition
None
Is empty
Is not empty
Text starts with
Text ends with
Text is exactly
Date is
Date is before
Date is after
Greater than
Greater than or equal to
OK
Jones
Sorvino
Jardine
Andrews
Pencil
Binder
Pen
Pen Set

In Excel, click the dropdown, then **Filter...** then make sure only **(Blanks)** is checked. Click **OK**.

Excel:





You can then review a list of all the rows with blank cells in that column.

5. Select all these cells and delete the rows except the row of column headers.

6. Return to the **Filter by condition** and return it to **None**. In Excel, click **Clear Filter from 'Column'**.

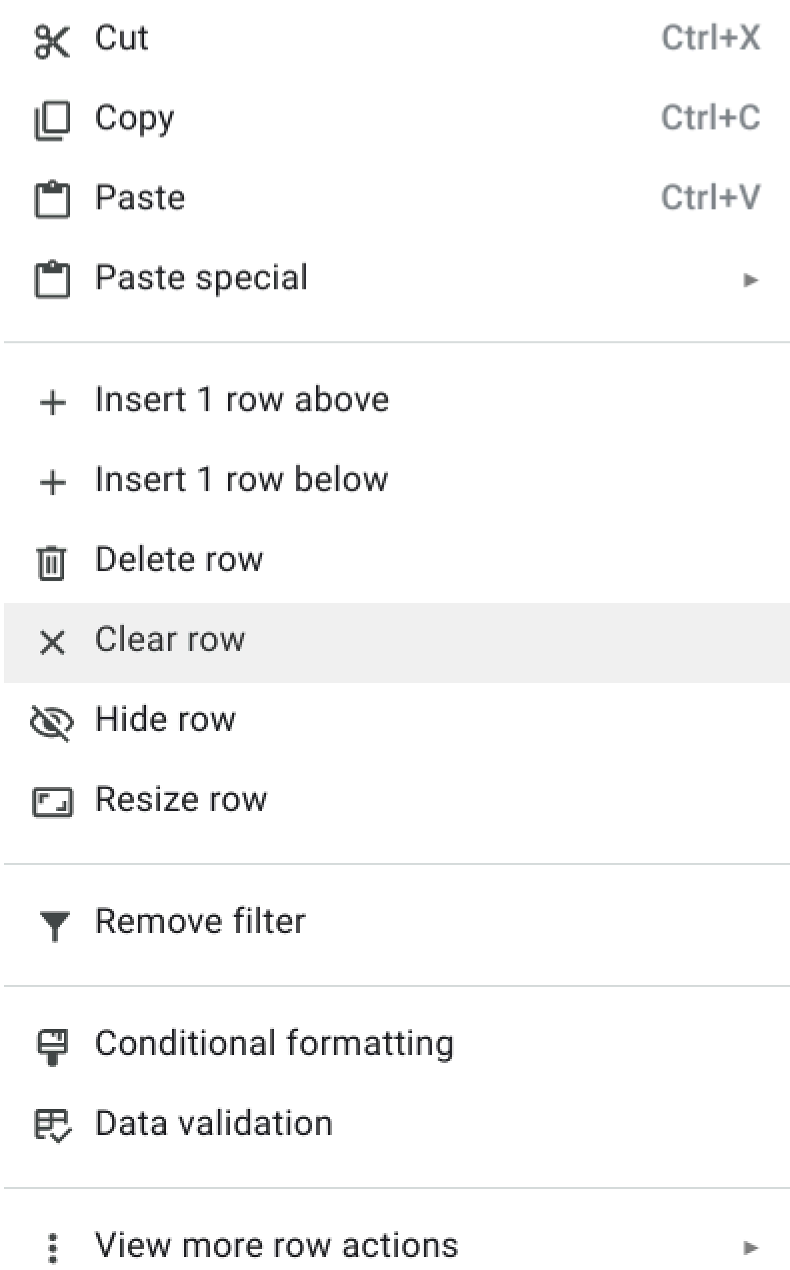
- **Note:** You will now notice that any row that had an empty cell in **Column A** will be removed (including the extra empty rows after the data).

7. Repeat this for **Columns B-H**.

8. **Note:** If you simply deleted the data from the row by tapping the backspace button, you will need to go a step further and *delete the empty row entirely* by left-clicking the row number located on the furthest left side of the screen.

	A	B	C	D	E	F	G	H
1	**Number**	**Order Date**	**Region**	**Name**	**Item**	**Units**	**Unit Cost**	**Total**
2								
3	1	1/6/19	East	Jones	Pencil	95	1.99	189.05
4	2	1/23/19	Central	Kivell	Binder	50	19.99	999.5

9. Next, right click on the highlighted row to call up the drop down window, and select the **Delete row** option.



10. Continue to do this same operation for the remaining empty rows in the data set.

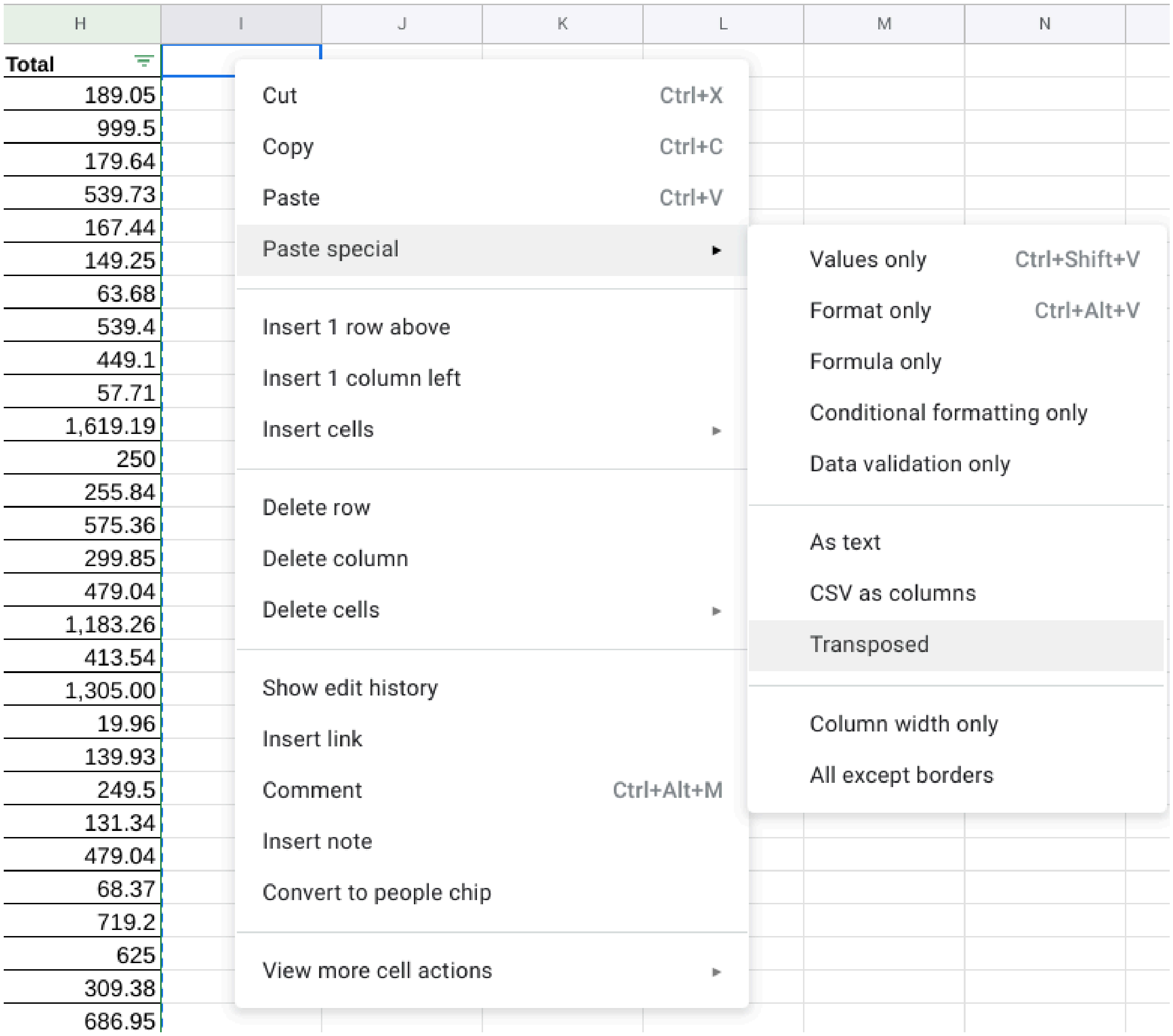
All the rows that had blank cells are now removed from the spreadsheet.

The second technique you will practice will help you convert the data from the current long format (more rows than columns) to the wide format (more columns than rows). This action is called **transposing.** To transpose your data:

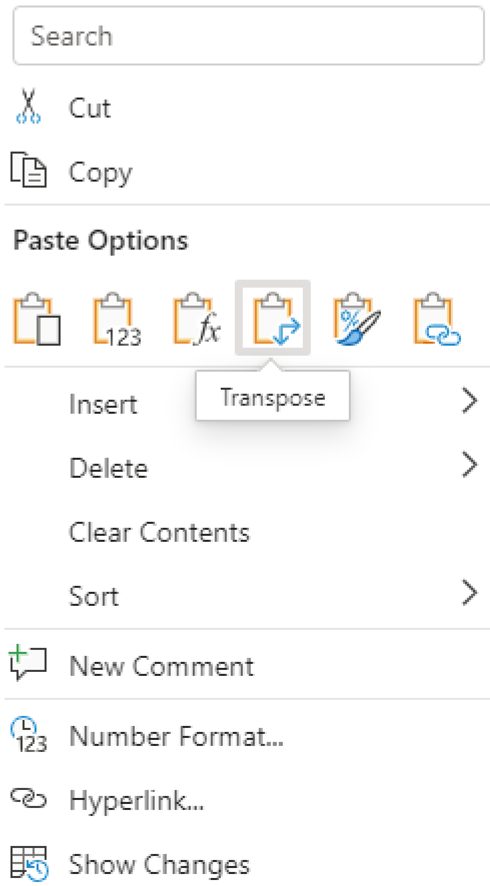
1. Highlight and copy the data that you want to transpose including the column labels. You can do this by highlighting **Columns A-H**. In Excel, highlight only the relevant cells (**A1-H45**) instead of the headers.

2. Right-click on **cell I1**. This is where you want the transposed data to start.

3. Hover over **Paste Special** from the right-click menu. Select the **Transposed** option. In Excel, select the **Transpose** icon under the paste options.

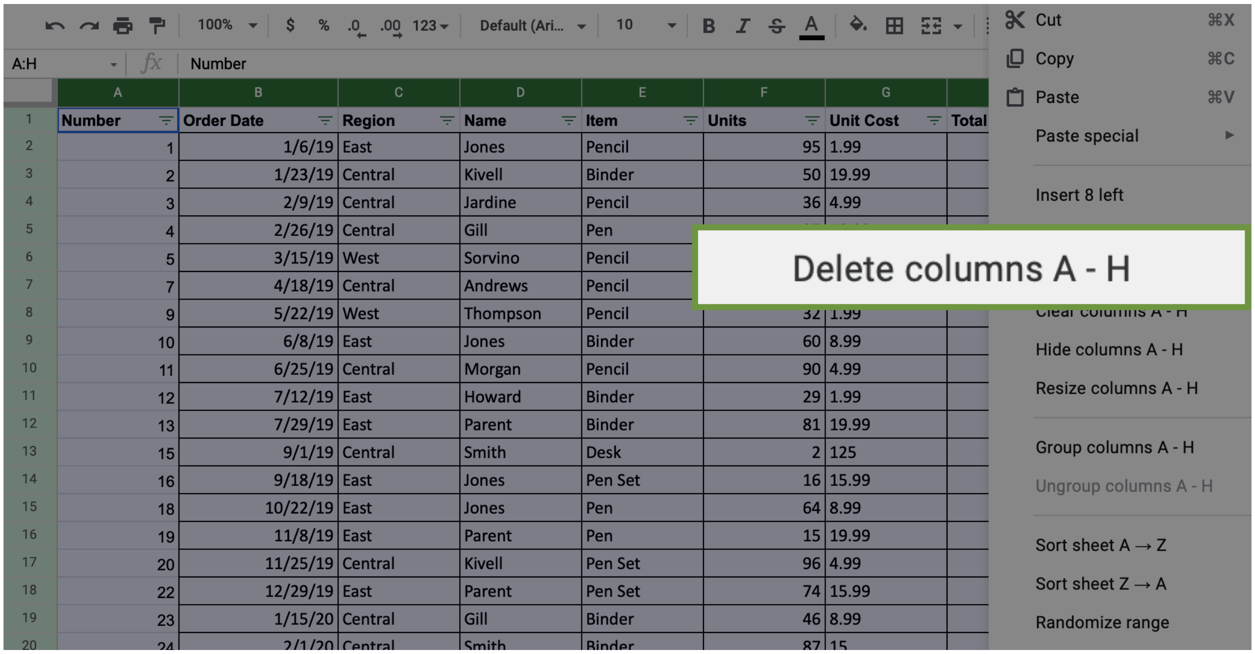

H
I
J
K
L
M
N
Total
189.05
999.5
179.64
539.73
167.44
149.25
63.68
539.4
449.1
57.71
1,619.19
250
255.84
575.36
299.85
479.04
1,183.26
413.54
1,305.00
19.96
139.93
249.5
131.34
479.04
68.37
719.2
625
309.38
686.95
Cut Ctrl+X
Copy Ctrl+C
Paste Ctrl+V
Paste special
Insert 1 row above
Insert 1 column left
Insert cells
Delete row
Delete column
Delete cells
Show edit history
Insert link
Comment Ctrl+Alt+M
Insert note
Convert to people chip
View more cell actions
Values only Ctrl+Shift+V
Format only Ctrl+Alt+V
Formula only
Conditional formatting only
Data validation only
As text
CSV as columns
Transposed
Column width only
All except borders

Excel:

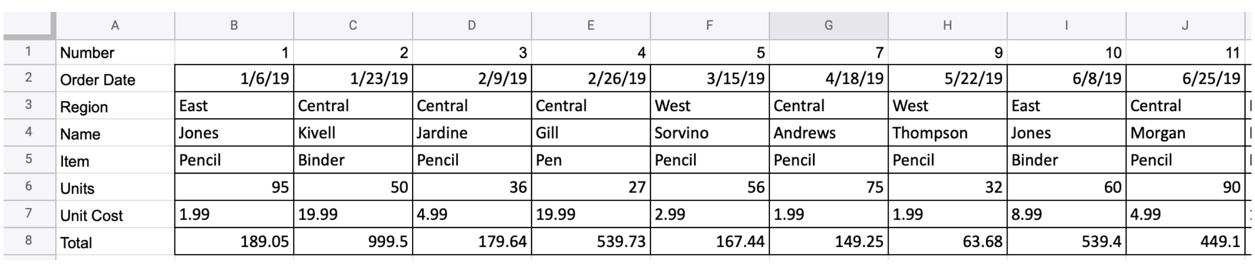


You should now find the data transformed into the new wide format. At this point, you should remove the original long data from the spreadsheet.

4. Delete the previous long data. The easiest way to do this is to click on **Column A**, so the entire column is highlighted. Then, hold down the **Shift** key and click on **Column H**. You should find these columns highlighted. Right-click on the highlighted area and select **Delete Columns A - H**.



Your screen should now appear like this:

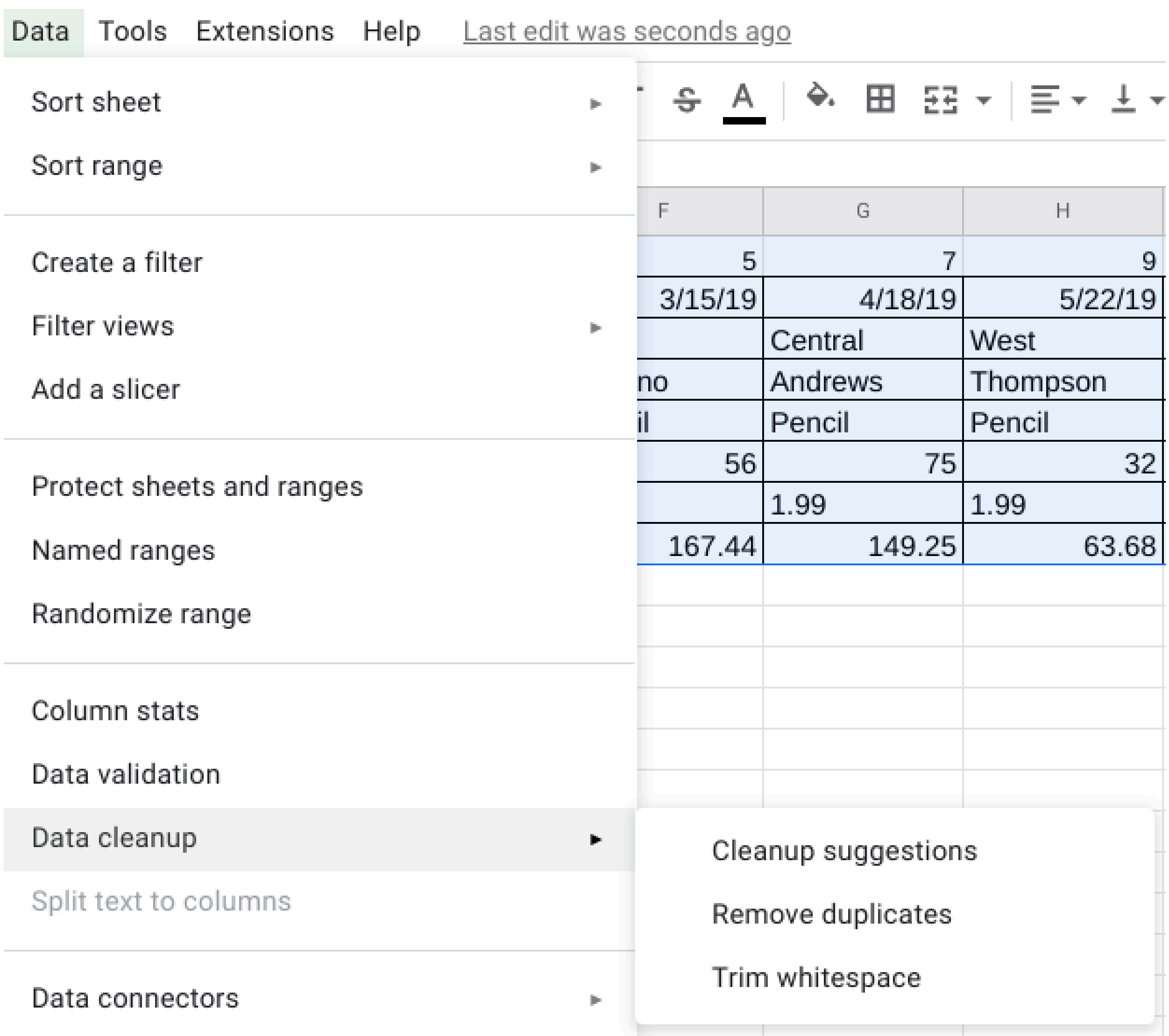


	A	B	C	D	E	F	G	H	I	J
1	Number	1	2	3	4	5	7	9	10	11
2	Order Date	1/6/19	1/23/19	2/9/19	2/26/19	3/15/19	4/18/19	5/22/19	6/8/19	6/25/19
3	Region	East	Central	Central	Central	West	Central	West	East	Central
4	Name	Jones	Kivell	Jardine	Gill	Sorvino	Andrews	Thompson	Jones	Morgan
5	Item	Pencil	Binder	Pencil	Pen	Pencil	Pencil	Pencil	Binder	Pencil
6	Units	95	50	36	27	56	75	32	60	90
7	Unit Cost	1.99	19.99	4.99	19.99	2.99	1.99	1.99	8.99	4.99
8	Total	189.05	999.5	179.64	539.73	167.44	149.25	63.68	539.4	449.1

Now that you have transposed the data, eliminate the extra spaces in the values of the cells.

1. Highlight the data in the spreadsheet.

2. Click on the **Data** tab, then hover over **Data cleanup** and select **Trim whitespace**.



In Excel, you can use the TRIM command to get rid of white spaces. In any space beneath your data (such as cell **A10**), enter `=TRIM(A1)`. Then, drag the auto-fill square handle at the bottom right corner of the cell and pull it downward to call the rest of the data in the column without the white spaces.

Note: It's important to carry out the `TRIM` formula for each column individually, as it may affect certain types of cell values (like dates) undesirably. Depending on the particular formatting of your column, you could decide on an appropriate workaround, such as first converting the column's values to strings. **In the case of this activity, you do not need to remove white spaces from the date values.**

For additional understanding of using the TRIM formula in Excel, please explore the Microsoft Excel guide on using the formula correctly.

Now all the extra spaces in the cells have been removed.

Next, you'll process string data. The easiest way to clean up string data will depend on the spreadsheet program you are using. If you are using Excel, you'll use a simple formula. If you are using Google Sheets, you can use an Add-on to do this with a few clicks. Follow the steps in the relevant section below.

Microsoft Excel

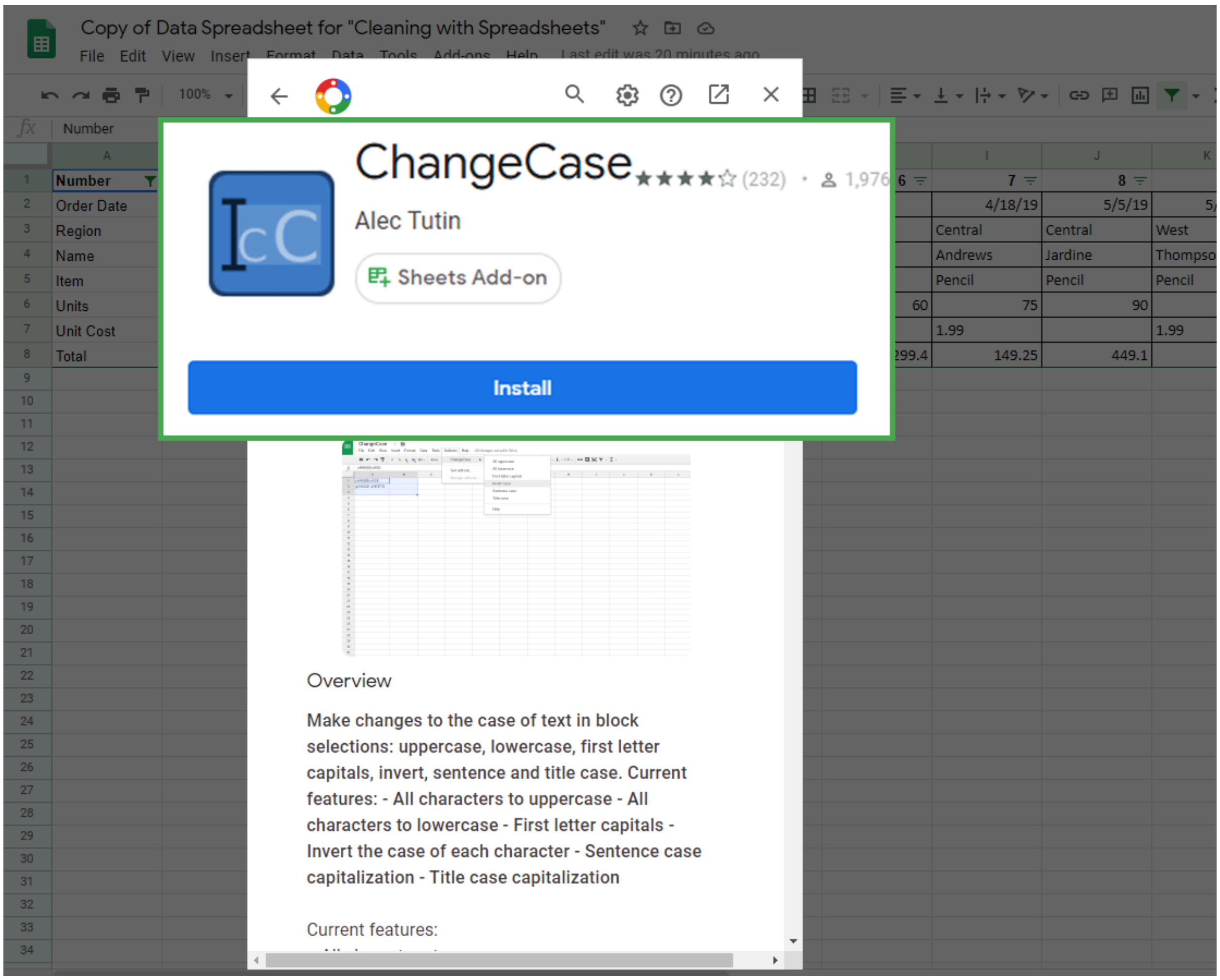
If you are using Microsoft Excel, this documentation explains how to use a formula to change the case of a text string. Follow these instructions to clean the string text and then move on to the confirmation and reflection section of this activity.

Google sheets

If you're completing this exercise using Google Sheets, you'll need to install an add-in that will give you the functionality needed to easily clean string data and change cases.

Google Sheets Add-on Instructions:

1. Click on the **Add-Ons** option at the top of Google Sheets.
2. Click on **Get add-ons**.
3. Search for **ChangeCase**. It should appear like this:



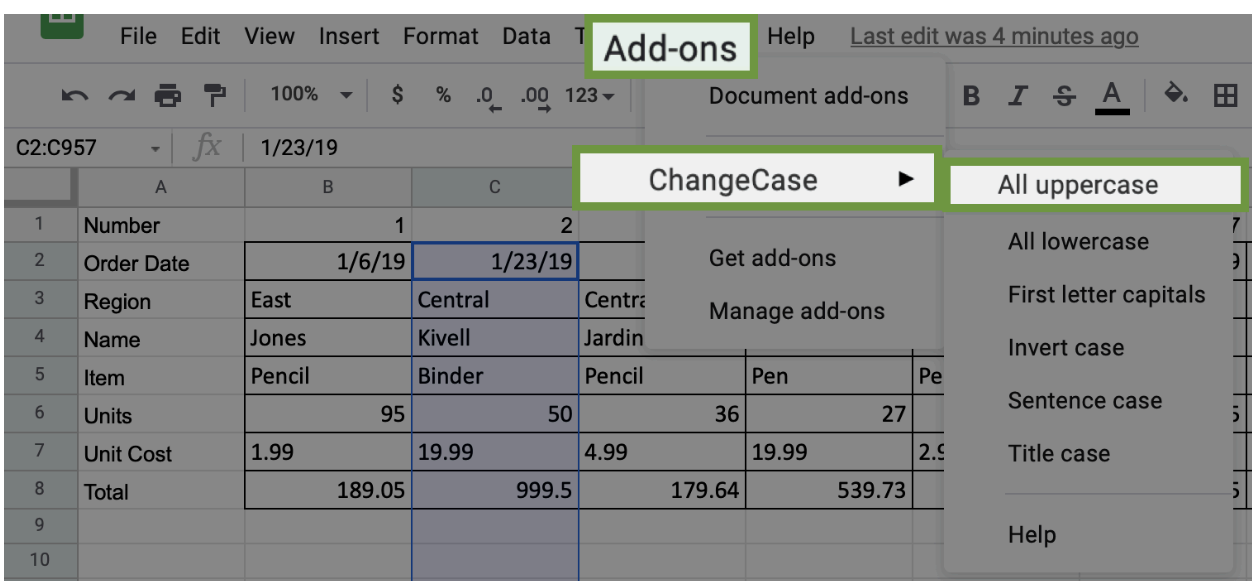
4. Click on **Install** to install the dd-on. It may ask you to login or verify the installation permissions.

Once you have installed the add-on successfully, you can access it by clicking on the **Add-ons** menu again.

Now, you can change the case of text data that shows up. To change the text in Column C to all uppercase:

1. Click on **Column C**. Be sure to deselect the column header, unless you want to change the case of that as well (which you don't).

2. Click on the **Add-Ons** tab and select **ChangeCase**. Select the option **All uppercase**. Notice the other options that you could have chosen if needed.



If you want to clear the formatting for any or all cells, you can find the command in the **Format** tab. To clear formatting:

1. Select the data for which you want to delete the formatting. In this case, highlight all the data in the spreadsheet by clicking and dragging over **Rows 1-8.**

2. Click the **Format** tab and select the **Clear Formatting** option.

In Excel, go to the **Home** tab, then hover over **Clear** and select **Clear Formats**.

You will notice that all the cells have had their formatting removed.

Be sure to save a copy of the spreadsheet template you used to complete this activity. You can use it for further practice or to help you work through your thought processes for similar tasks in a future data analyst role.

1.

Question 1

Reflection

Review the final product of the spreadsheet you cleaned during this activity. Which of the following is the rightmost column?

Column Z

Column AZ

Column AS

Column AA

Correct

In the final product of this activity, the rightmost column is Column AS. You are able to find this information after you properly transpose the data. Going forward, you can apply what you learned about data cleaning and transposing to work with data in the future.

2.

Question 2

In this activity, you practiced cleaning and transposing data. In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- What was the most challenging part of cleaning the data?
- Why is cleaning and transposing data important for data analysis?
- If you had to clean this data again, what would you do differently? Why?

My Answer:

I think that I didn't face any challenges through this exercise, but it was quite interesting for me. But if you tell me how challenging that was, the uppercase part was very time-consuming. Cleaning is the most vital part for the analysis; without cleaning, we can't get an appropriate result. Transposing is also very important because we can transform the data row to column or vice versa when we need it; we can utilize this very beautifully. Yeah, I do definitely do the different approach, because this time we have done it for practicing purposes, and in this approach we didn't get any insight, but next time if I do, then I will definitely do it for getting a proper insight.

Feedback

Congratulations on completing this hands-on activity! In this activity, you cleaned and transposed data on a spreadsheet. A good response would include that cleaning is a fundamental step in data science as it greatly increases the integrity of the data.

Good data science results rely heavily on the reliability of the data. Data analysts clean data to make it more accurate and reliable. This is important for making sure that the projects you will work on as a data analyst are completed properly.